

Hands in Bloom: An Auditory-Accessible Immersive Experience Based on Chinese Sign Language Recognition and Gesture Tracking

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Figure 1: (a)MR Gesture Interaction (b)Natural Signs Scene (c)Shukong Signs Scene (d)Ideographic Signs Scene.

Abstract

While gesture-based interaction in immersive experiences has become increasingly sophisticated, it primarily relies on natural gestures or user instructions delivered through on-site demonstrations. This work explores the integration of selected Chinese Sign Language (CSL) signs as defined interactive gestures embedded within immersive scenes to enable triggering and interaction. By incorporating CSL, the project not only expands the expressive possibilities of immersive interaction but also contributes to building a more inclusive and diverse experience space—one that creates a perceptible, barrier-free bridge between the Deaf and hearing communities.

CCS Concepts

• Human computer interaction → Interaction techniques.

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1 Introduction

Sign language bridges DHH and hearing communities, and recent systems now support bidirectional sign-text/speech (e.g., Signmaku, which stylizes signing as participatory “sign bullet comments” [Chen et al. 2024]). In interactive and immersive works, signing is coupled with light and vibration to convey musical affect—such as Ikigai Machine and Felt Sound (mapping ASL to low-frequency tactile rhythms) [McHugh et al. 2021]. As immersive experiences evolve, auditory-accessibility design continues to advance, and sign language is being integrated into a growing range of interactive scenarios.

Focusing on the cultural context of Chinese Sign Language (CSL), this paper proposes a method that categorizes CSL and maps it to actionable interaction gestures, and applies the method within immersive experiences.

2 OUR APPROACH

This project first categorizes Chinese Sign Language (CSL) based on gesture characteristics and cultural connotations, dividing it into three types: natural signs, which primarily consists of common gestures; Shukong signs, characterized by handwriting-like trajectories in the air; and ideographic hybrid signs, which draws on pictographic composition. On this basis, the signs are simplified and transposed into a set of gestures that are both system-recognizable and easy to learn.

The system is designed as an MR interactive experience, whose interaction logic corresponds to the three types of gestures. After selecting “Start” on the initial interface, users sequentially enter three MR scenes. In each scene, they follow gesture cues to perform the corresponding sign; once recognized, the system triggers semantic changes and feedback within the virtual environment. The current prototype includes 15 gestures, each paired with specific interactive elements and multimodal feedback (Fig. 2)

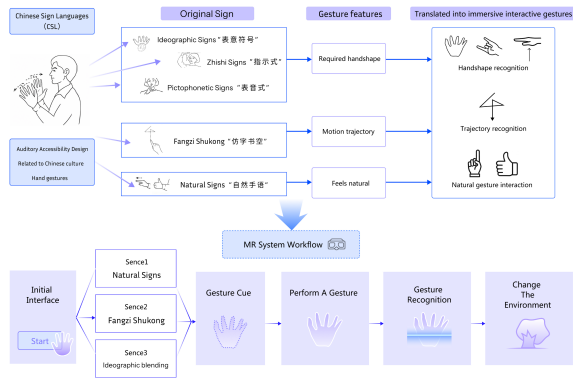


Figure 2: Sign-to-Gesture Translation and System Workflow

2.1 Natural Signs–Guided Spatial Interaction

The participant locates the virtual companion within the environment and interacts through natural signs to advance the experience. As a communicative gesture with social functionality, natural signs combines approachable linguistic expression with dynamic characteristics, effectively capturing attention while serving as a medium for spatial guidance and information transmission.

At the start of the experience, the virtual companion appears successively in different spatial positions, demonstrating natural signs such as “hello” and “continue” to direct the participant’s gaze and prompt corresponding actions. The participant follows the companion’s real-time three-dimensional gesture demonstrations to perform two gesture inputs. Once the gestures are successfully recognized, the system uses them as triggers for the companion to fade out, and the scene transitions to the next stage. (Fig3.a).

2.2 Multiple Mappings of Shukong Signs

Scene 2 Shukong signs as the primary interaction method. These signs imitate the writing process and stroke order of Chinese characters, and are commonly used to express simple words—particularly numbers.

After entering the experience, a writing trajectory guide appears in front of the participant, who follows the prompt to perform in-air gestures representing “hundred,” “thousand,” and “.” The system visualizes the writing path in real time using particle trails that follow the participant’s index finger. Once the gestures are recognized, the system sequentially executes the transformation from gesture-character-number, mapping the number to a corresponding quantity of water droplets. Through different “watering” interactions, participants trigger the growth of a virtual sapling. This gesture–character–number–feedback mapping chain concretizes abstract information, enhances comprehensibility, and allows both non-signers and non-Chinese participants to intuitively grasp its meaning.(Fig3.b)

2.3 Scene-Based Ideographic Blending

Scene 3, uses ideographic signs as the main interaction mode. These signs visually represent the shape or features of real-world objects,

making them intuitive and easy to understand. Participants perform gestures for “flower,” “grass,” and “mountain,” which generate corresponding 3D models.

In addition to model generation, this scene incorporates phonetic and deictic signs for further editing. Deictic signs, often used to express position, direction, or comparative attributes, are represented here by the contrasting gestures “high” and “low,” which allow users to adjust model height and form. For color control, phonetic signs are used—specifically “yellow,” “blue,” and “purple”—enabling users to intuitively change model colors through gesture. Two additional ideographic signs, “cut” and “restore,” are introduced to allow modifications and reversals of model form, empowering users to more flexibly engage in the creation and re-creation of the immersive environment.(Fig3.c)

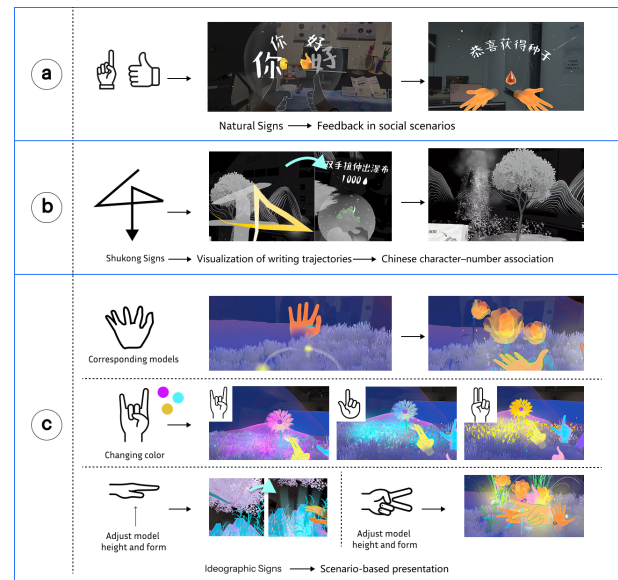


Figure 3: Scene Interaction Gestures and Feedback

Limitations include difficulties in distinguishing similar signs and recognizing dynamic gestures. Future work will enhance dynamic gesture capabilities and integrate CSL with XR for practical communication scenarios.

Acknowledgments

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